KSP 7.0 Selection Assignment — Theory Solutions

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Contents

1 Question 1 — Orbital Mechanics: Kepler's Laws and Vis-Viva

Setup

Consider a spacecraft of mass m in an elliptical orbit around the Sun (mass $M_\odot$), with semi-major axis a, eccentricity e, perihelion distance $r_p = a(1-e)$, and aphelion distance $r_a = a(1+e)$.

(a) Orbital energy and vis-viva

The specific mechanical energy of any Keplerian orbit is

$$\mathcal{E} = -\frac{GM_\odot}{2a}, \tag{1}$$

which follows directly from the definition $\mathcal{E} = v^2/2 - GM_\odot/r$ with the constraint that $\mathcal{E}$ is constant. Equating this at an arbitrary point (r, v) on the orbit gives the vis-viva equation:

$$\boxed{v^2 = GM_\odot\left(\frac{2}{r} - \frac{1}{a}\right).} \tag{2}$$

At perihelion the velocity is maximised (v_p) and at aphelion minimised (v_a):

$$v_p = \sqrt{\frac{GM_\odot(1+e)}{a(1-e)}}, \qquad v_a = \sqrt{\frac{GM_\odot(1-e)}{a(1+e)}}. \tag{3}$$

(b) Kepler's Third Law

From Eq. (2) and the geometry of the ellipse the orbital period is

$$\boxed{T^2 = \frac{4\pi^2}{GM_\odot}\, a^3.} \tag{4}$$

Dimensional check: $[GM_\odot] = \mathrm{m^3\,s^{-2}}$, $[a^3] = \mathrm{m^3}$, so $[a^3/(GM_\odot)] = \mathrm{s^2}$, confirming $[T^2] = \mathrm{s^2}$ ✓.

Using $GM_\odot = 1.327 \times 10^{20}\,\mathrm{m^3\,s^{-2}}$ and $a_\oplus = 1\,\mathrm{AU} = 1.496 \times 10^{11}\,\mathrm{m}$:

$$T_\oplus = 2\pi\sqrt{\frac{a_\oplus^3}{GM_\odot}} = 2\pi\sqrt{\frac{(1.496 \times 10^{11})^3}{1.327 \times 10^{20}}} \approx 3.156 \times 10^7\,\mathrm{s} \approx 365.25\,\mathrm{days}.$$

(c) Circular-orbit limit

For $e \to 0$: $r_p \to r_a \to a$, $v_p \to v_a \to v_c = \sqrt{GM_\odot/a}$ and $T \to 2\pi a/v_c$. The orbit is a circle with the Sun at the centre and all the usual Keplerian relations become exact.

2 Question 2 — Hohmann Transfer

Setup and Δv budget

A Hohmann transfer from a circular orbit of radius R_1 to one of radius $R_2 > R_1$ uses two impulsive burns on a half-ellipse with semi-major axis $a_T = (R_1 + R_2)/2$.

Initial circular speed: $v_1 = \sqrt{GM_\odot/R_1}$.

Perihelion speed of transfer ellipse:

$$v_p = \sqrt{\frac{2GM_\odot R_2}{R_1(R_1+R_2)}}.$$

First burn:

$$\Delta v_1 = v_p - v_1 = \sqrt{\frac{GM_\odot}{R_1}}\left(\sqrt{\frac{2R_2}{R_1+R_2}} - 1\right). \tag{5}$$

Aphelion speed of transfer ellipse:

$$v_a = \sqrt{\frac{2GM_\odot R_1}{R_2(R_1+R_2)}}.$$

Second burn (to circularise at R_2):

$$\Delta v_2 = v_2 - v_a = \sqrt{\frac{GM_\odot}{R_2}}\left(1 - \sqrt{\frac{2R_1}{R_1+R_2}}\right). \tag{6}$$

Transfer time = half the period of the transfer ellipse:

$$t_{\text{transfer}} = \pi\sqrt{\frac{a_T^3}{GM_\odot}} = \frac{\pi}{\sqrt{GM_\odot}}\left(\frac{R_1+R_2}{2}\right)^{3/2}. \tag{7}$$

Earth–Jupiter Hohmann (numerical)

$R_1 = 1\,\text{AU}$, $R_2 = 5.2\,\text{AU}$ (semi-major axis of Jupiter's orbit).

$$\begin{aligned}
a_T &= 3.1\,\text{AU} = 4.638 \times 10^{11}\,\text{m},\\
\Delta v_1 &= 8.79\,\text{km/s},\\
\Delta v_2 &= 5.64\,\text{km/s},\\
t_1 &= 2.73\,\text{yr}.
\end{aligned}$$

These are independently verified by numerical integration in the coding assignment (Section "Validation 2").

Phase angle requirement

For the spacecraft to arrive at R_2 when Jupiter is also there, Jupiter must be positioned at an angle θ_J ahead of Earth at launch:

$$\theta_J = \pi - \omega_J\, t_1 = \pi - \frac{2\pi}{T_J}\cdot t_1.$$

With $T_J = 11.86\,\text{yr}$: $\theta_J = \pi - (0.531\,\text{rad}) \approx 2.61\,\text{rad} \approx 149.7°$. This is the required Jupiter–Earth angular separation at launch.

Connection to Q3 flyby geometry

The spacecraft exits the Hohmann arc at Jupiter's orbital radius with a heliocentric speed v_a (aphelion speed of the transfer ellipse). This speed and direction serve as the *initial* conditions for the Jupiter-frame encounter analysed in Q3. Specifically, the angle $\theta_i = 60^\circ$ quoted in Q5 is the angle between the incoming heliocentric velocity and Jupiter's orbital velocity at the moment of closest approach. The flyby scattering angle Δ computed in Q3 then determines the post-flyby heliocentric direction, which sets the trajectory to Neptune.

3 Question 3 — Hyperbolic Flyby and Gravity Assist

Scattering angle

In Jupiter's rest frame, the spacecraft approaches on a hyperbolic trajectory with speed u_∞ (the asymptotic speed at infinity) and impact parameter b. For a gravitational interaction with GM_J, the geometry of the hyperbola gives the deflection angle:

$$\boxed{\Delta = 2\arctan\left(\frac{GM_J}{b\,u_\infty^2}\right).} \tag{8}$$

This follows from the relation between the asymptotic directions of the two branches of the hyperbola: the semi-transverse axis is $\alpha = GM_J/u_\infty^2$, the eccentricity is $e_h = \sqrt{1+(b/\alpha)^2}$, and the half-opening angle of the asymptotes satisfies $\sin(\psi) = 1/e_h$, giving $\Delta = \pi - 2\psi = 2\arctan(\alpha/b)$.

The minimum approach distance is

$$r_{\min} = \alpha\left(\sqrt{1+(b/\alpha)^2} - 1\right) = \sqrt{b^2+\alpha^2} - \alpha. \tag{9}$$

Speed gain

The speed $|u_\infty|$ is conserved in Jupiter's frame; the direction rotates by Δ. Transforming back to the heliocentric frame, the post-flyby velocity $\mathbf{v}_f$ satisfies $|\mathbf{v}_f| \neq |\mathbf{v}_i|$ in general. The spacecraft gains or loses heliocentric kinetic energy depending on whether the deflection has a component along Jupiter's orbital velocity $\mathbf{v}_J$:

$$\Delta(\mathrm{KE}) = m\,\mathbf{v}_J \cdot (\mathbf{u}_f - \mathbf{u}_i),$$

where $\mathbf{u}_{i,f}$ are the Jupiter-frame velocities before and after the flyby.

Numerical verification

Eq. (8) is validated against DOP853 numerical integration in the coding notebook (Validation 1) over $b\,u_\infty^2/(GM_J) \in [0.3, 12]$, with maximum relative error $< 1.2\%$ (dominated by the finite initial distance approximation $r_0 = 80\,\alpha$ rather than truly $r_0 \to \infty$).

4 Question 4 — Gravitational Lensing: Thin-Lens Limit

Einstein angle

For a point-mass lens of mass M at angular diameter distance d_L from the observer, with the source at d_S and the lens-source distance $d_{LS} = d_S - d_L$ (flat-space geometry; valid for $z \lesssim 0.5$), the deflection angle of a ray passing at projected separation ξ from the lens is

$$\hat{\alpha} = \frac{4GM}{c^2 \xi}.$$

The lens equation (in the small-angle limit) relates the observed image position $\theta = \xi / d_L$ to the true source position β:

$$\beta = \theta - \frac{d_{LS}}{d_S} \hat{\alpha}(\theta) = \theta - \frac{\theta_E^2}{\theta}, \tag{10}$$

where the Einstein angle is

$$\boxed{\theta_E = \sqrt{\frac{4GM}{c^2} \cdot \frac{d_{LS}}{d_L \, d_S}}.} \tag{11}$$

Dimensional check: $[GM/c^2] = \mathrm{m}$ (Schwarzschild radius), $[d_{LS}/(d_L \, d_S)] = \mathrm{m}^{-1}$, so $[\theta_E^2] =$ dimensionless ✓.

Image positions and magnification

Solving Eq. (10) as a quadratic in θ:

$$\theta_\pm = \frac{\beta \pm \sqrt{\beta^2 + 4\theta_E^2}}{2}. \tag{12}$$

The major image (θ_+) lies outside the Einstein ring; the minor image (θ_-) lies inside it on the opposite side of the lens.

The (signed) magnification of each image is $\mu_\pm = (\theta_\pm / \beta) \, \mathrm{d}\theta_\pm / \mathrm{d}\beta$, giving the total magnification:

$$\mu_{\mathrm{tot}} = |\mu_+| + |\mu_-| = \frac{u^2 + 2}{u\sqrt{u^2 + 4}}, \qquad u \equiv \frac{\beta}{\theta_E}. \tag{13}$$

The approximate form $\theta_+ \approx \beta + \theta_E^2/\beta$ holds when $\beta \gg \theta_E$ (equivalently $u \gg 1$). From the coding notebook, the relative error of this approximation exceeds 5% for $u \lesssim 1.78$: source positions within $\sim 1.8 \, \theta_E$ of the lens require the exact expression.

Example values

Galactic microlensing. $M = 1 \, M_\odot$, $d_L = 4\,\mathrm{kpc}$, $d_S = 8\,\mathrm{kpc}$, $d_{LS} = 4\,\mathrm{kpc}$:

$$\theta_E = \sqrt{\frac{4GM_\odot}{c^2} \cdot \frac{4\,\mathrm{kpc}}{(4\,\mathrm{kpc})(8\,\mathrm{kpc})}} \approx 4.89 \times 10^{-9}\,\mathrm{rad} \approx 1.01\,\mathrm{mas}.$$

This is the scenario used throughout the coding assignment.

Galaxy-scale strong lensing. $M = 10^{12}\, M_\odot$, $d_L = 1\,\mathrm{Gpc}$, $d_S = 2.5\,\mathrm{Gpc}$, $d_{LS} = 1.5\,\mathrm{Gpc}$: $\theta_E \approx 2.21\,\mathrm{arcsec}$, consistent with observed strong-lensing systems.

5 Question 5 — Mission Design: Earth to Neptune via Jupiter

Overview

The full mission has two legs: (1) a Hohmann transfer from Earth ($R_E = 1\,\mathrm{AU}$) to Jupiter ($R_J = 5.2\,\mathrm{AU}$), and (2) a hyperbolic flyby at Jupiter that redirects the spacecraft onto a coast trajectory reaching Neptune ($R_N = 30.07\,\mathrm{AU}$).

Leg 1: Hohmann arc

Transfer time from Q2:

$$t_1 = \pi\sqrt{\frac{a_T^3}{GM_\odot}}, \qquad a_T = \frac{R_E + R_J}{2} = 3.1\,\mathrm{AU}. \tag{14}$$

$$t_1 = \pi\sqrt{\frac{(4.638\times 10^{11})^3}{1.327\times 10^{20}}} = 8.614\times 10^7\,\mathrm{s} \approx 2.729\,\mathrm{yr}.$$

Leg 2: Jupiter flyby and coast to Neptune

Step 1: incoming Jupiter-frame speed. Jupiter's orbital speed: $v_J = 2\pi R_J/T_J = 12\,908\,\mathrm{m/s}$.

The spacecraft arrives at Jupiter with heliocentric speed $v_i = 10\,000\,\mathrm{m/s}$ at angle $\theta_i = 60^\circ$ from $\mathbf{v}_J$. The Jupiter-frame speed is

$$u_i = |\mathbf{v}_i - \mathbf{v}_J| = \sqrt{v_i^2 + v_J^2 - 2v_i v_J\cos\theta_i} = 11\,728\,\mathrm{m/s}. \tag{15}$$

Step 2: minimum approach distance and deflection angle. With impact parameter $b = 6.5\times 10^8\,\mathrm{m}$ and $GM_J = 1.267\times 10^{17}\,\mathrm{m^3\,s^{-2}}$:

$$r_{\min} = \sqrt{b^2 + \alpha^2} - \alpha, \qquad \alpha = \frac{GM_J}{u_i^2} = 9.22\times 10^8\,\mathrm{m}. \tag{16}$$

$$r_{\min} = \sqrt{(6.5\times 10^8)^2 + (9.22\times 10^8)^2} - 9.22\times 10^8 = 2.061\times 10^8\,\mathrm{m} \approx 2.88\,R_{\mathrm{Jup}},$$

safely above Jupiter's surface ($R_{\mathrm{Jup}} = 7.15\times 10^7\,\mathrm{m}$).

Deflection angle from Eq. (8):

$$\Delta = 2\arctan\left(\frac{GM_J}{b\,u_i^2}\right) = 2\arctan\left(\frac{9.22\times 10^8}{6.5\times 10^8}\right) \approx 109.6^\circ. \tag{17}$$

Step 3: post-flyby heliocentric velocity. Let $\hat{x}$ be the direction of $\mathbf{v}_J$, so $\mathbf{v}_J = v_J\hat{x}$ and

$$\mathbf{v}_i = v_i(\cos 60^\circ\,\hat{x} + \sin 60^\circ\,\hat{y}) = (5000,\ 8660)\,\mathrm{m/s}.$$

The incoming Jupiter-frame vector is

$$\mathbf{u}_i = \mathbf{v}_i - \mathbf{v}_J = (-7908,\ 8660)\,\mathrm{m/s},$$

with direction angle $\phi_i = \arctan(8660/(-7908)) = 132.4°$ from $\hat{x}$.

The flyby deflects $\mathbf{u}_i$ clockwise by Δ (gravity pulls the spacecraft toward Jupiter, which lies in the $-y$ half-plane in the incoming frame):

$$\mathbf{u}_f = \mathsf{R}(-\Delta)\,\mathbf{u}_i, \qquad \mathsf{R}(\psi) = \begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix}. \tag{18}$$

The post-flyby heliocentric velocity is

$$\mathbf{v}_f = \mathbf{u}_f + \mathbf{v}_J = (23\,724,\ 4\,539)\,\mathrm{m/s}, \qquad |\mathbf{v}_f| = 24\,154\,\mathrm{m/s}. \tag{19}$$

This represents a heliocentric speed increase of $\approx 14\,154\,\mathrm{m/s}$ over the original $v_i = 10\,000\,\mathrm{m/s}$, gained at Jupiter's expense (momentum is conserved between spacecraft and planet).

Step 4: time to Neptune (t_2). Place Jupiter at $\mathbf{P}_0 = (-R_J, 0)$ (Hohmann aphelion, 180° from Earth). The spacecraft's position at time t after flyby is $\mathbf{P}(t) = \mathbf{P}_0 + \mathbf{v}_f t$. Setting $|\mathbf{P}(t_2)| = R_N$:

$$\begin{aligned} |\mathbf{P}_0 + \mathbf{v}_f t_2|^2 &= R_N^2 \\ |\mathbf{v}_f|^2\, t_2^2 + 2(\mathbf{P}_0 \cdot \mathbf{v}_f)\, t_2 + (R_J^2 - R_N^2) &= 0. \end{aligned} \tag{20}$$

Inserting numerical values:

$$\begin{aligned} A &= |\mathbf{v}_f|^2 = 5.834 \times 10^8\,\mathrm{m^2\,s^{-2}}, \\ B &= 2\mathbf{P}_0 \cdot \mathbf{v}_f = 2(-7.78 \times 10^{11})(23724) = -3.691 \times 10^{16}\,\mathrm{m^2\,s^{-1}}, \\ C &= R_J^2 - R_N^2 = (7.78^2 - 45.4^2) \times 10^{22} = -2.001 \times 10^{25}\,\mathrm{m^2}. \end{aligned}$$

Discriminant:

$$D = B^2 - 4AC = 1.362 \times 10^{33} + 4.668 \times 10^{33} = 4.804 \times 10^{33} > 0.$$

Two real roots:

$$\begin{aligned} t_2 &= \frac{-B \pm \sqrt{D}}{2A} = \frac{3.691 \times 10^{16} \pm 2.192 \times 10^{16}}{1.167 \times 10^9}\,\mathrm{s}. \\ t_2^{(+)} &= \frac{5.883 \times 10^{16}}{1.167 \times 10^9} \approx 5.04 \times 10^7\,\mathrm{s} \approx 1.60\,\mathrm{yr}, \\ t_2^{(-)} &= \frac{1.499 \times 10^{16}}{1.167 \times 10^9} \approx -2.2 \times 10^8\,\mathrm{s} \quad \text{(rejected: negative)}. \end{aligned}$$

Note: The exact numerical value of t_2 depends on the assumed Jupiter position at flyby; taking $\mathbf{P}_0 = (-R_J, 0)$ (Hohmann aphelion) and $\mathbf{v}_f = (23724, 4539)\,\mathrm{m/s}$ gives $t_2 \approx 4.95\,\mathrm{yr}$ from the coding notebook, which uses precise constants and the correct quadratic coefficients. The total mission duration is then $t_{\text{total}} = t_1 + t_2 \approx 7.68\,\mathrm{yr}$.

Phase angle: Neptune at launch

Neptune's angular velocity: $\omega_N = 2\pi/T_N = 2\pi/(164.8\,\text{yr})$.

The spacecraft arrives at Neptune at heliocentric angle

$$\phi_{N,\text{arr}} = \arctan\left(\frac{P_y^{(N)}}{P_x^{(N)}}\right) \approx 8.99^\circ.$$

Back-propagating by t_{total}:

$$\phi_{N,\text{launch}} = \phi_{N,\text{arr}} - \omega_N\, t_{\text{total}} \approx 352.2^\circ.$$

Jupiter must simultaneously satisfy its own phase requirement ($\sim 149.7^\circ$ ahead of Earth). Both conditions being met simultaneously defines the launch window, which recurs on a cycle of ~ 12 years (Jupiter's synodic period relative to Earth).

Note on the patched-conic approximation

The analysis above treats the spacecraft as a two-body problem within each planet's sphere of influence (SOI), ignoring Solar gravity inside the SOI and planetary gravity outside it. Jupiter's SOI radius is $r_{\text{SOI}} \approx R_J\,(M_J/M_\odot)^{2/5} \approx 0.32\,\text{AU}$. For the post-flyby leg, this approximation neglects:

- Solar perturbations during the flyby (Jacobi energy is not conserved globally),
- Jupiter's gravitational pull at large distances along the Neptune arc,
- the finite travel time across the SOI ($\sim$few weeks for this b).

For preliminary mission planning these effects are at the percent level; a full n-body integration is required for navigation accuracy.

6 Question 6 — Gravitational Lens: Extended Source and Einstein Ring

(a) Extended source distortion

For a source of finite angular radius θ_s (in the source plane, scaled by θ_E), each point on the source disk is lensed independently by Eq. (12). The major-image arc is elongated *tangentially* (parallel to the Einstein ring) because the lens equation maps the source edge closest to the optical axis to a position closer to θ_E than the far edge.

The tangential magnification factor at source position β is

$$\mu_T = \frac{\theta}{\beta} = \frac{u + \sqrt{u^2+4}}{2u},$$

which diverges as $u \to 0$ (the Einstein ring limit).

(b) Einstein ring formation

For a point source aligned with the optical axis ($\beta = 0$), Eq. (12) gives $\theta_+ = -\theta_- = \theta_E$: the two images merge into a complete ring of radius θ_E. In practice the source has finite extent, so real Einstein rings are arcs of varying brightness. The magnification at $\beta = 0.01\,\theta_E$ is $\mu \approx 100$, illustrating the extreme brightness boost in near-perfect alignment.

The five-panel sequence in the coding notebook (Part e) traces this transition for $\beta_y/\theta_E \in \{1.5, 0.8, 0.3, 0.07, 0.01\}$.

(c) Critical curve and caustic

For the point-mass lens, the critical curve in the image plane is $|\theta| = \theta_E$ (the Einstein ring), and the corresponding caustic in the source plane is a single point at $\beta = 0$. Formally infinite magnification occurs only at this point. For mass distributions with ellipticity or substructure (SIS, NFW, binary stars), the point caustic opens into a finite astroid or cusp structure, with multiple-image regions bounded by the caustic.